

Genome Mining and Characterization of a Type III Polyketide Synthase Biosynthetic Gene Cluster (Region 8) in *Bacillus subtilis* Using antiSMASH and Comparative Bioinformatics.

Tools used:

- **Proksee**
- **antiSMASH**
- **MIBiG**
- **BLASTP**
- **Phylogeny**

***Bacillus subtilis* Genome**



Proksee

**(Genome visualization &
Basic annotation)**



antiSMASH

(Biosynthetic Gene Cluster Prediction)



Region 8 Selection

(Type III PKS)



MIBiG Comparison



Core Biosynthetic Gene

Identification



**Protein Sequence
Extraction**



BLASTP

(Homology Search)



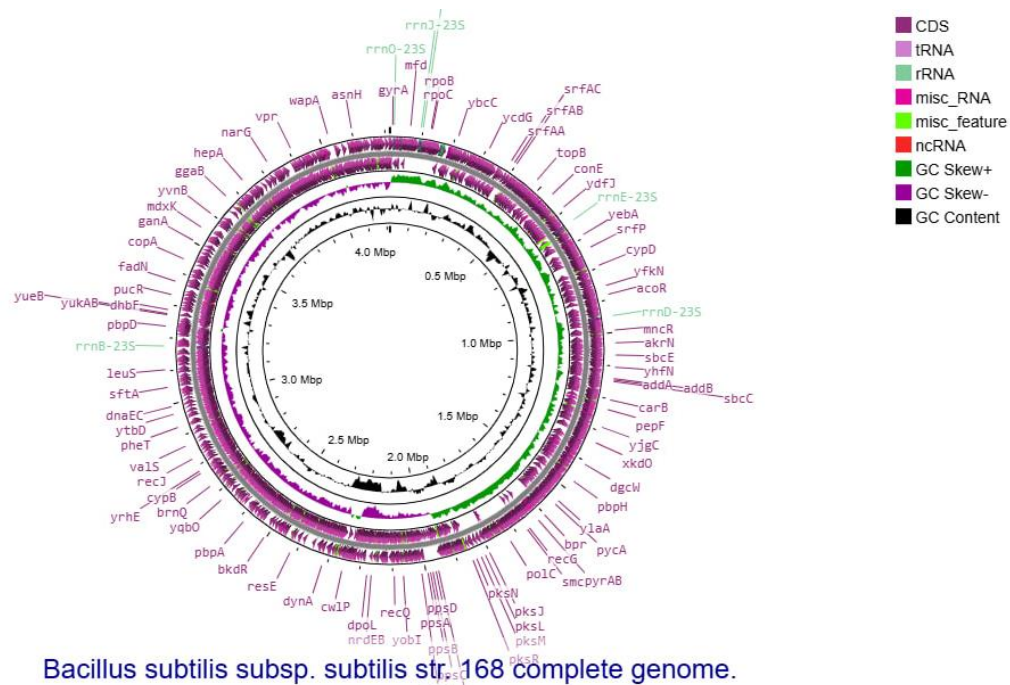
Multiple Sequence Alignment

(MAFFT)



Phylogenetic Analysis

(Phylogeny.fr)



Circular genome map of *Bacillus subtilis* generated using Proksee. The map displays coding sequences (CDSs), GC content, GC skew, and genomic features used as the starting point for biosynthetic gene cluster analysis.

Select genomic region: Overview 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9 1.10 1.11 1.12 1.13 1.14 1.15

Identified secondary metabolite regions using strictness 'relaxed'

NC_000964.3 (Bacillus subtilis subsp. subtilis str. 168)

Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 1	saccharopeptide ↗ , ranthipeptide ↗	203,941	226,894	High	sporulation killing factor ↗	ribosomal:unmodified
Region 2	NRPS ↗	356,968	422,359	High	surfactin ↗	NRPS:Type I
Region 3	terpene ↗	1,149,922	1,170,746	High		
Region 4	transAT-PKS ↗ , PKS-like ↗ , T3PKS ↗ , NRPS ↗	1,763,763	1,878,521	High	bacillaene ↗	NRPS:Type I+PKS:Type I
Region 5	NRPS ↗ , betalactone ↗	1,940,198	2,017,957	High	fengycin ↗	NRPS:Type I
Region 6	terpene ↗	2,092,168	2,114,066	High		
Region 7	glycocin ↗	2,259,521	2,279,691	High	sublancin 168 ↗	ribosomal:RIPP:Lanthipeptide
Region 8	T3PKS ↗	2,296,956	2,338,053	Low	1-carbapen-2-em-3-carboxylic acid ↗	other:other
Region 9	terpene-precursor ↗	2,515,789	2,536,679	High		
Region 10	terpene-precursor ↗ , NRP-metallophore ↗ , NRPS ↗	3,246,008	3,312,296	High	bacillibactin ↗	NRPS:Type I
Region 11	CDPS ↗	3,593,821	3,614,567	High	pulcherriminic acid ↗	other:other
Region 12	saccharopeptide ↗	3,826,058	3,847,669	High	subtilosin A ↗	ribosomal:RIPP:Thiopeptide
Region 13	other ↗	3,850,668	3,892,086	High	bacilysin ↗	other:other
Region 14	RRE-containing ↗	4,088,150	4,108,419			
Region 15	epipeptide ↗	4,115,742	4,137,440			

antiSMASH analysis identified 15 biosynthetic gene clusters in *Bacillus subtilis*. Region 8 was selected for further analysis due to its low similarity to known clusters


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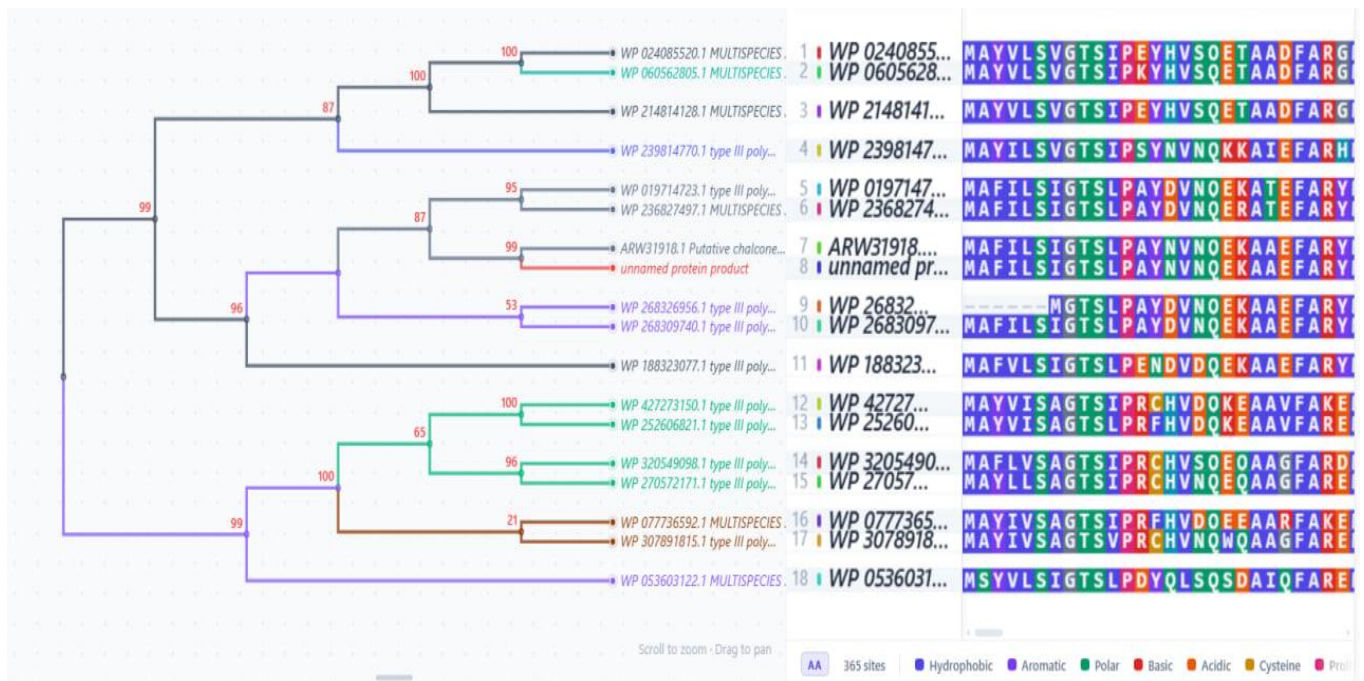
CLUSTAL format alignment by MAFFT (v7.511)

unnamed      MAFILSIGTSLPAYNVNQEKAAEFARYMFQHSFKDIDRLLSSFKNQGIHSRQFVKPIEWY
ARW31918.1   MAFILSIGTSLPAYNVNQEKAAEFARYMFQHSFKDIDRLLSSFKNQGIHSRQFVKPIEWY
WP_2368274   MAFILSIGTSLPAYDVNQERATEFARYMFQHSFKDIDRLLTSFKNGQIHSRQFVKPIEWY
WP_0197147   MAFILSIGTSLPAYDVNQEKAATEFARYMFQHSFKDIDRLLTSFKNGQIHSRQFVKPIEWY
WP_2683097   MAFILSIGTSLPAYDVNQEKAAEFARYMFQHSFKDIDRLLTSFKNGQIQTRQFVKPIEWY
WP_2683269   -----MGTSLPAYDVNQEKAAEFARYMFQHSFKDIDRLLTSFKNGQIQSRQFVKPIEWY
WP_1883230   MAFVLSIGTSLPENDVDQEKAAEFARYMFQHSFKDIERLLTSFKNGQIESRQFVRPIEWY
WP_2398147   MAYILSVGTSIPSYNVNQQKAIEFARHMFQDSFKDIERLLKAFENGIEIESRQFVRPIEWY
WP_2148141   MAYVLSVGTSIPEYHVSQETAADFARGMFAQSFKDIDRLLKVFNQNGEIESRQFVRPIEWY
WP_0605628   MAYVLSVGTSIPKYHVSQETAADFARGMFADSFKDIDRLLKAFQNGEIESRQFVRPIEWY
WP_0240855   MAYVLSVGTSIPEYHVSQETAADFARGMFADSFKDIDRLLKAFQNGEIESRQFVRPIEWY
WP_0777365   MAYIVSAGTSIPRFHVDQEEAARFAKEMFKHSFKDIDRLLASFKNGEIESRQFVMPLEWY
WP_3078918   MAYIVSAGTSVPRCHVNQWQAAGFAREMFKDSFKDIDRLLASFKNGEIESRQFAMPLEWY
WP_2705721   MAYLLSAGTSIPRCHVNQEQAAGFAREMFKDSFKDIDRLLASFKNGEIESRQFVMPLEWY
WP_3205490   MAFLVSAGTSIPRCHVSEQEAAGFARDMFKDSFRDIDRLLASFKNGEIQSRQFVMPIEWY
WP_2526068   MAYVISAGTSLPRFHVDQKEAAVFAREMFKDSFKDIDRLLAAFQNGEIASRQFVMPPLDWY
WP_4272731   MAYVISAGTSIPRCHVDQKEAAVFAKEMFKDSFKDIDRLLAAFQNGEIASRQFVMPPLDWY
WP_0536031   MSYVLSIGTSLPDYQLSQSDAIQFAREMFKDSFDIERLLKSFTNGDIESRQFVKPLEWY

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Multiple sequence alignment of the predicted Region 8 Type III PKS protein with representative homologs. Conserved amino acid residues are highlighted, indicating a high level of sequence conservation among bacterial Type III polyketide synthases.



Phylogenetic analysis of the predicted Type III polyketide synthase protein showed that the query sequence clustered with Bacillus type III PKS homologs. The closest homolog was **WP_236827497.1** with high sequence similarity, suggesting that the predicted protein belongs to the bacterial **Type III polyketide synthase** family.